

From Helices to Supercoiling: Local and Global Geometry of DNA

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Abstract

Classical invariants such as curvature and torsion completely describe the local behavior of curves, they fail to capture global coiling in closed structures. In this paper, we begin with the standard helix, whose constant curvature and torsion make it an example in differential geometry. We then construct closed curves via toroidal helices and explore iterated coiling using Frenet frames. This leads to the Călugăreanu–White–Fuller formula, which decomposes the linking number into twist and writhe. Through qualitative analysis of parameter variations and handedness, we demonstrate how global coiling emerges and connect these ideas to DNA supercoiling.

Introduction

Helices are fundamental objects in differential geometry and appear naturally in physical and biological systems, most notably in the structure of DNA. To what extent can the geometry of DNA be understood using classical invariants such as curvature and torsion?

While these invariants describe the local behavior of curves, DNA often exists in closed, supercoiled configurations where global geometric effects become important. This paper explores the transition from local to global geometry by studying helices, their generalizations, and the introduction of topological invariants.

The Standard Helix

Consider the standard helix

$$\mathbf{r}(t) = (a \cos t, a \sin t, bt),$$

where $a, b > 0$.

The curvature and torsion are given by

$$\kappa = \frac{a}{a^2 + b^2}, \quad \tau = \frac{b}{a^2 + b^2}.$$

Both are constant, implying that the helix has uniform bending and twisting. Thus, the helix is geometrically homogeneous and serves as a fundamental example of a curve completely described by local invariants.

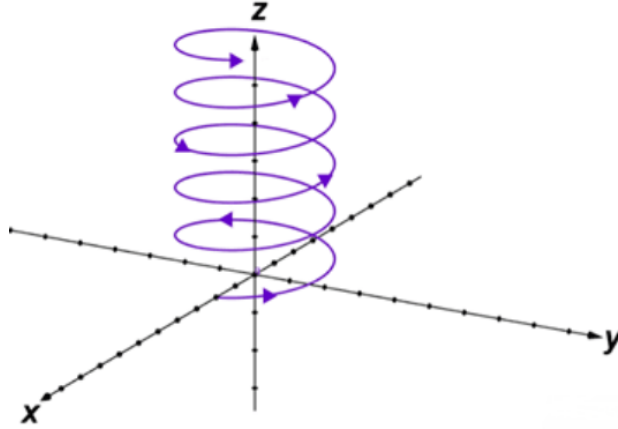


Figure 1: Standard Helix

Limitations of Local Geometry

Curvature and torsion are local invariants: they describe how a curve behaves infinitesimally. However, they do not capture how different parts of a curve interact globally.

In particular, when curves are closed, global properties such as coiling and entanglement arise. These cannot be detected purely through local quantities, motivating the need for additional invariants.

Toroidal Helices

To model closed helical structures, consider the toroidal helix:

$$\mathbf{r}(t) = ((R + a \cos(nt)) \cos t, (R + a \cos(nt)) \sin t, a \sin(nt)),$$

where $R > a > 0$ and $n \in \mathbb{Z}$.

This curve is closed and represents a helix wrapped around a circle. Unlike the standard helix, its curvature and torsion are no longer constant.

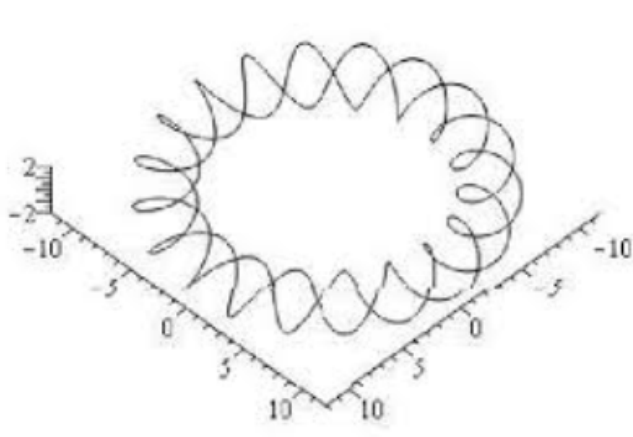


Figure 2: Toroidal Helix

Observations:

- Increasing n increases the number of oscillations.
- Increasing a increases the amplitude of coiling.

These changes significantly affect the global geometry, even when local variations are small.

Helices on Helices

A further generalization can be constructed using the Frenet frame. Given a curve $\gamma(t)$ with normal $N(t)$ and binormal $B(t)$, define:

$$\Gamma(t) = \gamma(t) + \epsilon(\cos(nt)N(t) + \sin(nt)B(t)).$$

This produces a curve that winds around γ , creating a “helix on a helix.” Iterating this process produces increasingly complex coiled structures.

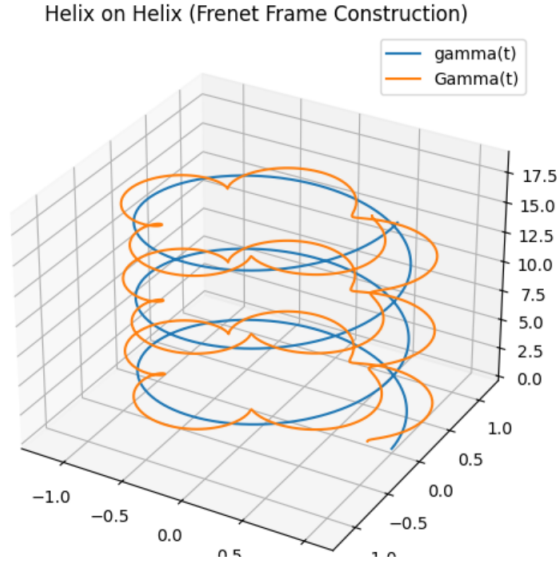


Figure 3: Helix on Helix

Such constructions illustrate how geometric complexity increases through successive layers of coiling.

Global Invariants: Twist, Writhe, and Link

To describe closed curves, we introduce the Călugăreanu–White–Fuller formula:

$$Lk = Tw + Wr$$

- **Linking number (Lk):** a topological invariant counting how two curves wind around each other.
- **Twist (Tw):** measures local rotation of a ribbon around its centerline.

- **Writhe (Wr):** measures global coiling of the curve in space.

Unlike curvature and torsion, writhe captures global interactions along the curve.

Qualitative Analysis

Effect of Oscillation Number

As n increases, the number of oscillations increases, leading to more crossings in projections of the curve. This suggests that writhe increases with n .

Effect of Handedness

Changing the sign of the sine term reverses the handedness of the coiling.

Observation:

- Consistent handedness leads to reinforcement of coiling.
- Opposite handedness leads to partial cancellation.

This behavior reflects the interaction between twist and writhe in the relation $Lk = Tw + Wr$.
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Connection to DNA Supercoiling

DNA consists of a double helix that is often constrained in closed configurations. When the molecule is over- or under-twisted, it cannot simply relax locally. Instead, twist is converted into writhe, producing supercoiling.

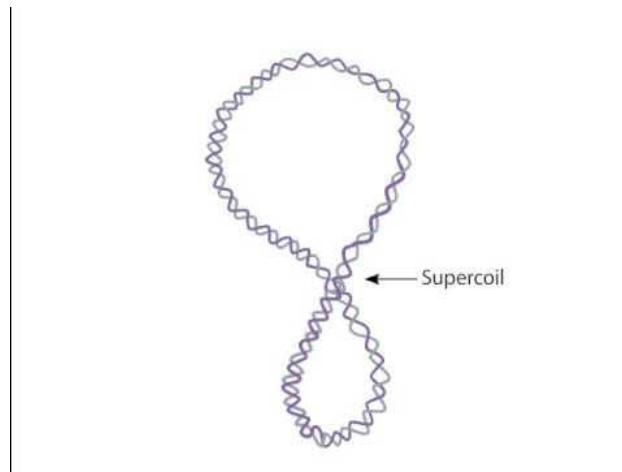


Figure 4: DNA structure with supercoiling and double helix

- **Positive supercoiling:** twist and writhe reinforce.
- **Negative supercoiling:** twist and writhe oppose.

Thus, global geometric effects play a crucial role in the structure and behavior of DNA.

Further Remarks

The Fáry–Milnor theorem states that a knotted curve must have total curvature at least 4π . This highlights the interplay between local geometric quantities and global topology, reinforcing the necessity of considering global invariants.

Conclusion

While curvature and torsion provide a complete description of local geometry, they are insufficient for understanding closed curves with complex coiling. The introduction of global invariants such as writhe and twist allows for a deeper understanding of these structures. Through the study of toroidal helices and iterated coiling, we see how local and global geometry interact, providing insight into the phenomenon of DNA supercoiling.